Computer Organization and Architecture (CSL251)
LAB-2
2025-26 W

RISC-V Assembly Programming Assignment

Instructions:

1. Write separate RISC-V assembly programs for each question.
2. Use the RISC-V GNU toolchain (riscv64-unknown-elf-gcc) to assemble and link your code.
3. Run your program using the Spike RISC-V simulator.
4. After execution, display relevant register values using Spike commands such as:
    a. reg 0 (to view all registers)
    b. reg 0 a0 (to view result register)
5. Take clear screenshots showing:
    a. Successful compilation
    b. Program execution in Spike
    c. Final output register values
6. Upload the following for submission under a single zip file:
    a. All assembly source files (.s files)
    b. Screenshots of execution and output
7. Clearly name your files using the format:
   Q1_swap.s, Q2_add.s, Q3_prime.s, etc.

Question 1: Swapping Two Numbers

Write a RISC-V assembly program to swap two 32-bit word values stored in memory. Load the values into registers t0 and t1 using the lw instruction, then swap their contents.

Expected Output

The value originally loaded into t0 should be stored in t1

The value originally loaded into t1 should be stored in t0

Question 2: Arithmetic Operation

Write a RISC-V assembly program to compute the following expression:

$$t4 = (15 \times 7) + (20 \div 4) - 3^2$$

Store the final result in register t4.

Use immediate arithmetic and register operations as required.

Question 3: Prime Number Test

Write a RISC-V assembly program to test whether a number stored in register a1 is prime or not.

Store the Boolean result in register a0.

a0 = 1 if the number is prime.

a0 = 0 if the number is not prime.

Question 4: Perfect Square Check

Write a RISC-V assembly program to check whether a natural number stored in register a1 is a perfect square or not.

Store the Boolean result in register a0.

a0 = 1 if the number is a perfect square.

a0 = 0 otherwise.

Question 5: Factorial Using Recursion

Write a RISC-V assembly program to compute the factorial of a number stored in register a1 using recursion.

Store the final result in register a0.

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